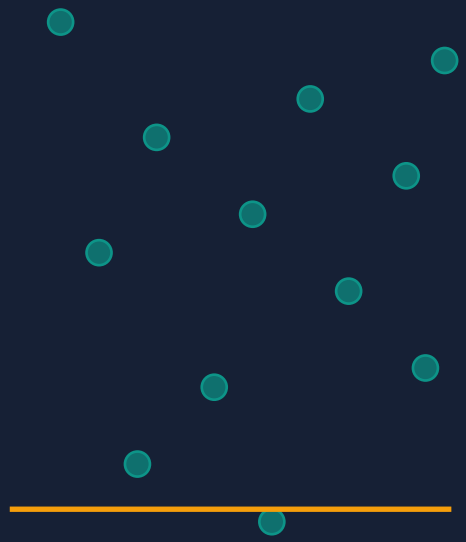
Linear Regression

Theory, Assumptions & RStudio

Based on Andy Field – Discovering Statistics Using R



Muhammad adil zainal abidin

What is Linear Regression?

"A statistical method that models the relationship between one or more predictor variables and a continuous outcome variable by fitting a straight line."

For prediction and control for confounder , quantifying the relationship -Simple and multiple (crude and adjusted)

Outcome (Y)	Predictor (X)	The Line
The variable we want to predict or explain (dependent variable)	The variable(s) used to predict the outcome (independent variable)	Best-fit line minimising the sum of squared residuals (OLS method) Other method Maximum Likelihood Estimation (MLE) for logistic regression

The Regression Equation:

$$\hat{Y}_i = b_0 + b_1 X_i + \varepsilon_i$$

b_0 = intercept b_1 = slope/coefficient ε_i = error/residual

Field (2013): Discovering Statistics Using R, p. 183

Theory: How OLS Works

Ordinary Least Squares (OLS)

- Each data point has a residual: the vertical distance between the observed (Y) and predicted ($\hat{Y}$) value
- OLS finds the line that minimises the Sum of Squared Residuals (SSR)
- $SSR = \Sigma(Y_i - \hat{Y}_i)^2 \rightarrow$ minimised
- Squaring ensures negative and positive errors don't cancel out
- The resulting b_0 and b_1 are the 'best' linear estimates

Model Fit: R^2 & F-ratio

R^2

Proportion of variance in Y explained by the model. $R^2 = 0.60 \rightarrow$ model explains 60% of variance in Y.

Adjusted R^2

Penalises for extra predictors. Preferred when comparing models (Field, 2013).

F-ratio

Tests whether the model significantly improves prediction over the mean alone. $F = MS_m / MS_r$

β (Beta)

Standardised coefficients allow comparison of predictor importance across different scales.

Field (2013), pp. 184–205

Assumptions of Linear Regression

1

Linearity

Relationship between X and Y must be linear. Check: scatterplot of X vs Y; residuals vs fitted plot.

2

Independence

Observations must be independent of each other. Check: study design; Durbin-Watson test (≈2 = no autocorrelation).

3

Homoscedasticity

Residuals should have constant variance across levels of X. Check: residuals vs fitted plot — no funnel shape.

4

Normality of Residuals

Residuals should be normally distributed. Check: Q-Q plot, histogram of residuals, Shapiro-Wilk test.

5

No Multicollinearity

Predictors should not be too highly correlated (multiple regression). Check: VIF < 10; Tolerance > 0.1 (Field, 2013).

6

No Influential Outliers

Extreme cases unduly influence the model. Check: Cook's Distance > 1 suggests influential cases (Field, 2013).

Field (2013), Chapter 7 – All assumptions must be met for valid inference.

Running Linear Regression in RStudio (Step 1–3)

STEP 01

Load & Inspect Your Data

```
# Load data
library(tidyverse)
data <- read.csv("mydata.csv")

# Inspect
head(data)
str(data)
summary(data)
```

Ensure outcome is continuous and predictors are correctly coded.

STEP 02

Visualise the Relationship

```
# Scatterplot
ggplot(data, aes(x = predictor,
                 y = outcome)) +
  geom_point(colour = "steelblue") +
  geom_smooth(method = "lm",
              se = TRUE) +
  theme_minimal()
```

Look for linear trend before fitting the model.

Assumption 1
Linearity

STEP 03

Fit the Regression Model

```
# Simple linear regression
model <- lm(outcome ~ predictor,
            data = data)

# View full output
summary(model)
```

lm() = linear model. Outcome ~ Predictor syntax follows R formula notation.

Field (2013), pp. 211–220 | Code adapted for R / tidyverse

Interpreting the summary() Output

```
Coefficients:

            Estimate  Std.Error  t value   Pr(>|t|)
(Intercept)  10.543     1.234     8.543    <0.001 ***
predictor     0.621     0.087     7.138    <0.001 ***

---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

Residual standard error: 3.21 on 98 df
Multiple R-squared:  0.612
Adjusted R-squared:  0.608
F-statistic: 51.95 on 1 and 98 DF
p-value: < 2.2e-16
```

Intercept (b_0)

Predicted Y when X = 0

Slope ($b_1 = 0.621$)

For each 1-unit ↑ in X, Y increases by 0.621

p-value < .001

Predictor is statistically significant

$R^2 = 0.612$

Model explains 61.2% of variance in outcome

F-statistic

Overall model is significant ($p < .001$)

Field (2013), pp. 221–235

Checking Assumptions in RStudio (Step 4)

What to Look For:

```
# ── Dubin watson test ──────────────────
dwtest(model)

# ── Diagnostic Plots ──────────────────
# Base R: 4 plots in one window
par(mfrow = c(2,2))
plot(model)

# ── Homoscedasticity ──────────────────
library(lmtest)
bptest(model)  # Breusch-Pagan test

# ── Normality of Residuals ────────────
shapiro.test(residuals(model))

# ── Multicollinearity (if multiple X) ──
library(car)
vif(model)       # VIF < 10 = acceptable
tolerance <- 1/vif(model)  # > 0.1

# ── Influential Cases ─────────────────
cooksD <- cooks.distance(model)
plot(cooksD, type="h")
abline(h = 1, col="red", lty=2)
```

Independence

Dubin Watson test – p value not significant and value ~ 2 (Assumption 2)

Residuals vs Fitted

Random scatter around zero → linearity & homoscedasticity OK (Assumption 3)

Scale-Location

Horizontal band → constant variance (homoscedasticity) (Assumption 3) p value should be not significant

Normal Q-Q Plot

Points on diagonal line → residuals are normally distributed (Assumption 4)

VIF / Tolerance

VIF < 10 and Tolerance > 0.1 → no multicollinearity (Assumption 5)

Cook's Distance

No points above 1 → no unduly influential cases (Assumption 6)

Field (2013), pp. 236–255 | car package by Fox & Weisberg (2019)

VIOLATION 1 LINEARITY

SITUATION

Look visually at scatter plot and residual plot
Scatter plot – DV vs IV
Residual plot – residual vs fitted values

Either transform

Or use non-linear regression

WHAT CAN YOU DO

- ✓ Use polynomial regression
- ✓ Use spline regression
- ✓ Use transformation (i.e log / square root)

```
poly_model <- lm(y ~ poly(x, 2))
plot(x, fitted(poly_model))
spline_model <- lm(y ~ ns(x, df=5))
plot(x, fitted(spline_model))
y_log <- log(y)
y_sqrt <- sqrt(y)
```

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

VIOLATION 2 INDEPENDENCE / AUTOCORRELATION

SITUATION

P Value significant or Dubin Watson value not ~ 2

Suggesting time series

Use time series model – ARIMA

WHAT CAN YOU DO

✓ Use time series model

ARIMA

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

VIOLATION 3 Homoscedasticity

SITUATION

residuals vs fitted plot — no funnel shape
Or Breusch pagan test p value significant

Either use robust standard error

Or Transform data

Or use Weighted Least Square (WLS) Method

WHAT CAN YOU DO

- ✓ Use robust standard error - Use the sandwich package to calculate heteroscedasticity-consistent (HC) standard errors and lmtest for testing.
  coeftest(model, vcov = vcovHC(model, type = "HC3"))
- ✓ Transform data by log transform the outcome - model_log <- lm(log(y) ~ x, data = dataset)
- ✓ WLS Method

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

VIOLATION 4 Residual normality

SITUATION

Q-Q plot not normal

If sample size is big , result is still robust

Check for outliers as well and remove them

Either transform the dependent variable

Or use robust regression

WHAT CAN YOU DO

✓ Transform the DV - model2 <- lm(log(y) ~ x, data = df)

✓ Robust regression - library(robustbase); robust_model <- lmrob(y ~ x, data = df)

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

VIOLATION 5 MULTICOLLINEARITY

SITUATION

2 or more predictor variables in a regression model are highly correlated, leading to unreliable estimates of the coefficients

Either drop one and re run

Or Ridge regression is a method by which we add a degree of bias to the regression estimates.

WHAT CAN YOU DO

- ✓ Drop the highly correlated predictors
- ✓ Regularization (ridge/lasso regression)

Repeat model without one of the correlated predictors

```
ridge_model <- glmnet(as.matrix(cbind(x1, x2, x3)), y, alpha=0)
summary(ridge_model)
plot(ridge_model)
coef(ridge_model, s = 0.1)
```

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

VIOLATION 6 OUTLIERS

SITUATION
Cook distance – if value >1 or 4/n

Check data entry first

Or data transformation of the dependent variable

Or delete the outliers

Or using robust regression

WHAT CAN YOU DO

Robust regression-
rlm() (MASS package): Uses Iteratively Reweighted Least Squares (IRLS).

```
library(MASS)
robust_model <- rlm(y ~ x, data = mydata)
summary(robust_model)
```

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

Linear regression - Moderator

To test moderation, we include an **interaction term**:

$$Y = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 (X \times M) + \epsilon$$

If **β3 is statistically significant**, moderation exists.

Check and visualize

```
model <- lm(Outcome ~ Predictor * Moderator, data = mydata)
summary(model)

library(interactions)
interact_plot(model, pred = Predictor, modx = Moderator)
```

Interpret

use sim_slopes from the interactions package
(model, pred = Predictor, modx = Moderator)

examine the effect of on at different levels of
(e.g., SD)

Interaction level

Categorical and categorical
Categorical and numeric
Numeric and numeric

Field (2013): Discovering Statistics Using R, p. 183

Linear regression – Mediator

Baron & Kenny Regression Approach (Classic Method)
In this classic method, mediation is tested using **three regression models**.

Step 1: Total Effect

$Y=\beta 0+\beta 1X$
If $\beta 1$ is significant → X affects Y.

Step 2: X → Mediator

$M=\beta 0+\beta 1X$
If significant → X influences mediator.

Step 3: Mediator → Outcome

$Y=\beta 0+\beta 1X+\beta 2M$
Interpretation:
β2 significant → mediator affects Y
If **β1 decreases** compared to Step 1 → mediation exists

The Regression Equation:

Full mediation - Effect of X on Y disappears after mediator added
Full mediation - Effect of X decreases but remains significant

Field (2013): Discovering Statistics Using R, p. 183

Mediator – modern approach using indirect effect

Modern mediation uses **indirect effect**:

$$Indirect\ Effect = a \times b$$

Where:
a = effect of X → M
b = effect of M → Y

Significance usually tested using:
Bootstrapping (preferred)
Sobel test

WHAT CAN YOU DO

- ✓ Bootstrapping - bootstrapping for mediation analysis is commonly done using packages such as mediation, lavaan, or boot. The easiest and most commonly used approach is the mediation package.
- ✓ Sobel test , using BDA package – significant p value indicated mediator

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

Reporting Results – APA Style (Field, 2013)

Example Write-Up

A simple linear regression was conducted to predict exam scores from revision hours. The assumptions of linearity, independence, homoscedasticity, and normality of residuals were all met. The model was statistically significant, $F(1, 98) = 51.95$, $p < .001$, $R^2 = .61$. Revision hours significantly predicted exam scores, $b = 0.62$, $t(98) = 7.14$, $p < .001$. For every additional hour of revision, exam scores increased by 0.62 marks on average.

Reporting Checklist

- ✓ State assumptions checked and results
- ✓ Report F-statistic, df, and p-value for overall model
- ✓ Report R^2 (and Adjusted R^2 for multiple regression)
- ✓ Report b (unstandardised) and β (standardised) coefficients
- ✓ Report t-value and p-value for each predictor
- ✓ Describe direction and magnitude of the effect in plain language

Field (2013), pp. 255–260 | APA Publication Manual, 7th Ed.

Regression table

Beta coefficient / OR

Standard error

t statistics

p value

Confidence interval

Table 3

Moderator Analysis: Types of Measurement and Study Year

confidence intervals in separate columns

Effect	Estimate	*SE*	95% CI		*p*
			LL	*UL*	
Fixed effects					
Intercept	.119	.040	.041	.198	.003
Creativity measurement[a]	.097	.028	.042	.153	.001
Academic achievement measurement[b]	–.039	.018	–.074	–.004	.03
Study year[c]	.0002	.001	–.001	.002	.76
Goal[d]	–.003	.029	–.060	.054	.91
Published[e]	.054	.030	–.005	.114	.07
Random effects					
Within-study variance	.009	.001	.008	.011	<.001
Between-study variance	.018	.003	.012	.023	<.001

Note. Number of studies = 120, number of effects = 782, total *N* = 52,578. CI = confidence interval; *LL* = lower limit; *UL* = upper limit.

[a]0 = self-report, 1 = test. [b]0 = test, 1 = grade point average. [c]Study year was grand centered.

[d]0 = other, 1 = yes. [e]0 = no, 1 = yes.

Table 7.16 Sample Regression Table, With Confidence Intervals in Brackets

Table 4

Regressions of Associations Between Marital Satisfaction and Average Levels of Marital Behavior

Variable	*B*	*SE*	*t*	*p*	95% CI
			Angry behavior		
Actor					
H → H	−98.90	40.20	−2.46	.016	[− 179.1, −18.7]
W → W	−.87.11	30.87	−2.82	.006	[−148.7, −25.6]
Partner					
W → H	−76.18	39.43	−1.93	.057	[−154.8, 2.4]
H → W	−91.80	38.16	−2.41	.019	[−167.9, −15.7]
			Disregard		
Actor					
H → H	−38.62	27.86	−1.39	.170	[−94.2, 16.9]
W → W	−47.54	26.99	−1.76	.082	[−101.4, 6.3]
Partner					
W → H	−82.81	32.01	−2.59	.012	[−146.6, −19.0]
H → W	−79.36	27.16	−2.92	.005	[−133.5, −25.2]
			Distancing		
Actor					
H → H	−47.42	24.72	−1.92	.059	[−96.7, 1.9]
W → W	3.04	23.48	0.13	.897	[−43.8, 49.8]
Partner					
W → H	−0.05	23.91	0.00	.998	[−47.7, 47.6]
H → W	−53.50	24.47	−2.19	.032	[−102.3, −4.7]

Note. CI = confidence interval; H → H = husband-as-actor effect on the husband's own marital satisfaction; W → W = wife-as-actor effect on the wife's own marital satisfaction; W → H = wife-as-partner effect on the husband's satisfaction; H → W = husband-as-partner effect on the wife's satisfaction.

square brackets around confidence intervals

Summary & Key Takeaways

1. Linear regression models a straight-line relationship between X and Y
2. OLS minimises the Sum of Squared Residuals to find the best-fit line
3. Always check the 6 assumptions before interpreting results
4. Use lm() in R; plot(model) gives 4 essential diagnostic plots
5. Report F, R^2, b coefficients, t-values and p-values in APA style
6. Field (2013) recommends checking Cook's D > 1 for influential cases

Questions & Discussion

Key Reference

Field, A. (2013).
Discovering Statistics Using R.
SAGE Publications.

Chapters 7–8
pp. 183–260